

JOANNA A. SANCHEZ

Major: Molecular & Cellular Biology | **Institution:** University of Illinois Urbana-Champaign (May 2026)

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Resilient and purpose-driven molecular and cellular biology student with a deep interest in immunology and computational biology

My academic journey has been defined by perseverance and a commitment to transforming challenges into opportunities for scientific growth and impact. I aim to advance research that bridges experimental immunology with computational science to uncover how the immune system responds to and is evaded by disease. I aspire to become a scientist who not only drives innovation at the interface of biology and technology but also broadens access and representation in biomedical research.

EDUCATION

5/2026	University of Illinois Urbana-Champaign , Illinois, USA
8/2023	<i>Undergraduate Student at the College of Liberal Arts & Sciences, School of Molecular & Cellular Biology</i> Major: Molecular and Cellular Biology
8/2022	Waubonsee Community College , Illinois, USA
8/2017	<i>Associate in Applied Science</i> Major: Biology

PROFESSIONAL APPOINTMENT

8/2024	Waubonsee Community College , Illinois, USA
3/2023	<i>Academic Support Paraprofessional Tutor (biology, anatomy)</i> ➤ Providing subject matter expertise and serving as models for students in the techniques, strategies, attitudes, and beliefs of successful learners ➤ Reviewing class material, discussing problems, reviewing tests, or working on solutions to problems ➤ Facilitating individual and small group tutoring sessions, and classroom visits

RESEARCH EXPERIENCE

present	Comparative Transcriptomics of the Drosophila Midgut: Regional Trypsin Specialization & Human Ortholog Mapping
1/2026	<i>Student Researcher (Advisor: Sheela Vemu, Ph.D.)</i> ➤ Conceived and executed an independent comparative transcriptomics project investigating regional specialization of the adult <i>Drosophila</i> midgut ➤ Built and documented a configurable Python/R bioinformatics workflow for RNA-seq analysis, enabling reproducible transcriptomic studies across multiple biological research questions ➤ Performed PCA, differential expression, and pathway enrichment analyses to characterize region-specific transcriptional programs across seven midgut compartments ➤ Identified posterior-enriched trypsin-family genes and linked them to human digestive-protease orthologs, demonstrating conservation of functional specialization across approximately 600 million years of evolution ➤ Presented results at the GRADS-4C undergraduate research symposium (June 2026)
2/2024	Computational Immunology (IEDB-based SARS-CoV-2 Epitope Analysis)
9/2022	<i>Student Researcher at Waubonsee Community College</i> ➤ Leading the study of the immunogenic regions on the spike glycoproteins of the SARS-CoV-2 variants using the web-based bioinformatics platform Immune Epitope Database (IEDB) ➤ Optimizing the target antigens' position by developing Python scripts, utilizing immune epitope prediction and analysis codes, statistical interpretation of data, and creating educational materials

- › Mentoring students for their honors project which uses IEDB:
 - › Jatniel Gomez (Student at Waubonsee Community College)
 - › Chris Salgado (Student at Waubonsee Community College)

HONORS AND AWARDS

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| 2026 | Helping Others' Personal Excellence Scholarship |
| 2023 | <i>University of Illinois Urbana-Champaign</i> <ul style="list-style-type: none"> › Awarded for demonstrated financial need and academic excellence |
| 2023 | NIIMBL eXperience
<i>The National Institute in Manufacturing Biopharmaceuticals</i> <ul style="list-style-type: none"> › Selected as one of ten students nationwide from U.S. community colleges |
| 2023 | First place at the Illinois Water Environment Association Student Poster Competition
<i>2023 Illinois Wastewater Professionals Conference</i> <ul style="list-style-type: none"> › Poster title: Mapping Immunogenic Regions in SARS-CoV-2 Variants to Understand PCR Assay Design Using Bioinformatics Tools › Newsletter |
| 2020 | Waubonsee Students Recognized on Fall 2020 President's List
<i>Waubonsee Community College</i> <ul style="list-style-type: none"> › Recognized for achieving a perfect 4.0 GPA |
| 2019 | The John J. Swalec, Jr. President's Achievement Award
<i>Waubonsee Community College</i> <ul style="list-style-type: none"> › Recognized for perseverance and excellence in achieving educational goals despite challenges |
| 2017 | Waubonsee Students Recognized on Fall 2017 President's List
<i>Waubonsee Community College</i> <ul style="list-style-type: none"> › Recognized for achieving a perfect 4.0 GPA |

CONFERENCES

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| 06/08/2026 | Trypsin Genes: How the Function of Trypsin Genes Is Mapped to Human Pancreatic Proteases
<i>Poster presentation at 3rd Annual Research Symposium (Genomic Research and Data Science Center for Computation and Cloud Computing) - Greensboro, NC</i> |
| 06/27/2023 | Mapped out immunogenic regions in SARS-CoV2 variants to understand vaccine design
<i>Poster presentation at the Envision the Next Bioscience Workforce Summit - Washington D.C</i> |
| 05/17/2023 | Training the Future of the Bioeconomy: Creating Accessible Research Experiences in Protein Engineering Using the Immune Epitope Database
<i>Poster presentation at PEGS Boston Summit 2023 Conference - Boston, MA</i> |
| 04/28/2023 | Mapping Immunogenic Regions in SARS-CoV-2 Variants to Understand Vaccine Design Using Bioinformatics Tools
<i>Podium Session at 2023 John Saragon Albazi Research and Creative Activities Student Symposium - Northeastern Illinois University, Chicago, IL</i> |
| 04/10/2023 | Mapping Immunogenic Regions in SARS-CoV-2 Variants to Understand Vaccine Design Using Bioinformatics Tools, (1st place winner)
<i>Student poster session at 2023 Illinois Wastewater Professionals Conference - Springfield, IL</i> |

PUBLICATIONS

- [1] Koudjiwan, P. *et al.*, "Predicting Neoantigens for Cancer Using Next-generation IEDB & CEDAR Tools", *Journal of Advanced Technological Education (J ATE)*, vol. 3, no. 2, Zenodo, Oct. 2024, pp. 47–56, doi:10.5281/zenodo.13989043
- [2] Sanchez Rocha, J. A. *et al.*, "Mapping Immunogenic Regions in Sars-cov-2 to Understand Vaccine Design Using Bioinformatics", *Journal of Advanced Technological Education (J ATE)*, vol. 2, no. 2, Zenodo, Aug. 2023, doi:10.5281/zenodo.8260361

ACTIVITIES

2024	Journal of Advanced Technological Education (J ATE)
2023	<i>NSF-Funded Undergraduate Research Experience (URE), InnovATEBIO Network</i> › Participated in hands-on biotechnology research projects aligned with industry needs, contributing to project design, data collection, and data analysis › Collaborated with faculty and peers to disseminate research outcomes through the <i>J ATE</i>

LAB COURSEWORK

Spring 2026	Eukaryotic Cell Biology Lab	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Spring 2025	Experimental Techniques in Cellular Biology	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Spring 2025	Organic Chemistry Lab	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Spring 2025	Physics E&M & Modern Lab	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Fall 2024	Physics Mechanism & Heat Lab	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Fall 2024	Molecular Genetic Lab	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Fall 2024	General Chemistry Lab II	UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN
Summer 2022	Anatomy and Physiology II Lab	WAUBONSEE COMMUNITY COLLEGE
Spring 2022	General Chemistry Lab	WAUBONSEE COMMUNITY COLLEGE
Spring 2020	Anatomy and Physiology I Lab	WAUBONSEE COMMUNITY COLLEGE
Fall 2019	Microbiology Lab	WAUBONSEE COMMUNITY COLLEGE

LEADERSHIP

2021	Student organization MEChA	WAUBONSEE COMMUNITY COLLEGE
2019	<i>President, Vice-president, Secretary</i> › Opened a chapter at Waubonsee that advocates for Latino students on campus › Provided resources and support for students' academic success	
2021	The Sideline Project	
2019	<i>Co-founder</i> › Conducted community events on mental health, academic success, and overall well-being › Collected data on available scholarships and opportunities for students	
2020	Student Diversity Committee	WAUBONSEE COMMUNITY COLLEGE
2019	<i>Student Leader</i> › Curating ideas to improve students' experience › Assisted in organizing student events for Latinx Heritage month	
2018	Education Awareness Subteam	
2017	<i>Student Leader</i> › Working in ensuring Waubonsee Community College took proactive action to protect DACA/Undocumented students › Provided interactive ally training for employees at the college, and students	

OUTREACH

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| 2020 | Good Morning Aurora <ul style="list-style-type: none"> ➤ Served as a special guest to speak about my advocacy work and share my story |
| 2018 | Supporting DACA and Undocumented Students Panelist <ul style="list-style-type: none"> ➤ Served as a panelist for high school educators to provide guidance in supporting DACA/Undocumented students |

SKILLS

Programming	R, Python, C++, Bash, HTML, TeX LaTeX, Git
Bioinformatics & Data Analysis	RNA-seq differential expression (DESeq2), pathway enrichment (KEGG/GO, clusterProfiler), Bioconductor, data visualization (ggplot2), cross-species ortholog mapping (FlyBase, Human Protein Atlas)
Software & Tools	Docker, R Markdown, YAML, pytest/CI, Visual Studio, Microsoft Word, Microsoft Excel, Photoshop, Google Suite
Languages	English (Native / Bilingual), Spanish (Native / Bilingual)